When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area *A* and a thickness *T* is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\text{DA}\,\Delta\,P}{T}$$

Equation 1

where *D* is the diffusion coefficient specific for the gas and ΔP is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration *C* of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\text{H}} P_{\text{gas}}$$

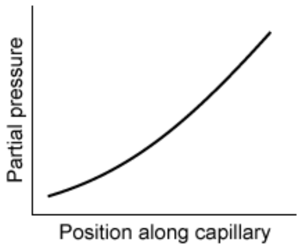
Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

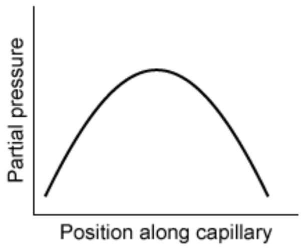
Which of the following graphs best depicts the anesthetic partial pressure along the length of the pulmonary capillaries when it is being administered? Capillaries are in contact with alveolar membranes along the entire length shown in the graphs. Assume there is no anesthetic in the blood when it enters the lungs.

○ A.



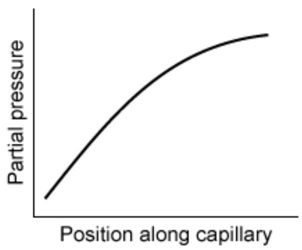
[19%]

○ B.



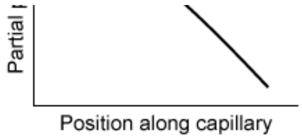
[10%]

✔ ◉ C.



[37%]

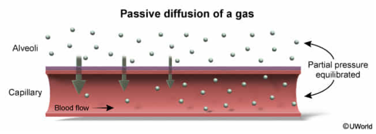
○ D.



[33%]

Correct
36% answered correctly

Explanation:



Gas exchange between the alveolar sacs and the pulmonary capillaries occurs due to **passive diffusion** and is **driven by partial pressure differentials** (difference) across the membrane from higher to lower partial pressures. The rate of diffusion (V_{gas}) of the anesthetic is described by the Fick law of diffusion (Equation 1) and is directly proportional to the partial pressure difference ΔP between the alveoli and the capillaries:

$$V_{\text{gas}} = \frac{\text{DA}\ \Delta\ P}{T}$$

When the anesthetic is being administered, its partial pressure is greater in the lungs, and therefore it will diffuse into the capillaries. The **capillary partial pressure will build up** along the length of the capillary as the blood flows through it. The alveolar partial pressure does not change significantly because the volume of gas in the lungs is much greater than the volume of gas in the capillaries.

As the partial pressure in the capillary increases, the partial pressure difference (ΔP) across the membrane driving diffusion will decrease. The rate of diffusion will decrease and will eventually reach zero when the capillary and alveolar **partial pressures are equilibrated**. This behavior is similar to that of oxygen exchange in the alveoli.

(Choices A and B) These graphs do not reflect the initial increase in partial pressure as the anesthetic diffuses into the capillaries and then the leveling off when the partial pressures in the lungs and capillaries reach equilibrium.

(Choice D) The partial pressure increases, not decreases, along the length of the capillary.

Educational objective:
The diffusion of a gas across a membrane is directly proportional to its partial pressure difference (Fick law of diffusion). Gases diffuse passively along the partial pressure gradient to reach equilibrium.

Time spent:0 sec
Subject:
Physics

QID:400655
System:
Fluids

When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area *A* and a thickness *T* is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\text{DA}\,\Delta\, P}{T}$$

Equation 1

where *D* is the diffusion coefficient specific for the gas and Δ*P* is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration *C* of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\text{H}} P_{\text{gas}}$$

Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

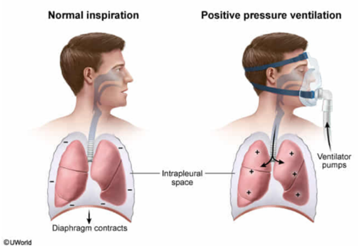
Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

A positive pressure mechanical ventilator most likely inflates the lungs by directly:

- A. increasing intrapleural pressure. [24%]
- B. decreasing intrapleural pressure. [34%]
- ✔ C. increasing alveolar pressure. [33%]
- D. decreasing alveolar pressure. [7%]

Correct
33% answered correctly

Explanation:



Respiration occurs in concert with lung volume changes and is driven by **pressure differentials** between the lungs (alveoli) and the pleural (chest) cavity. The change in pressures during inspiration is initiated by different means in normal and mechanical ventilation.

Normal inspiration is initiated by contraction of the diaphragm. As the diaphragm contracts, the volume of the pleural cavity increases and **intrapleural pressure (IPP) decreases**. The lungs expand to fill the pleural cavity and air flows into the lungs. Because normal inspiration is initiated by decreasing IPP, this pressure mechanism is referred to as *negative pressure* breathing.

According to the passage, general anesthetics have a paralyzing effect. Therefore, the diaphragm cannot initiate inspiration by decreasing IPP. Instead, the **positive pressure ventilator** directly pumps air into the lungs. This **increases alveolar pressure**, which causes the lungs to inflate.

(Choice A) An increased IPP would push against the lungs and oppose inhalation. In addition, the pleural cavity is separated from the external environment, and therefore a noninvasive mechanical ventilator would be unable to directly affect its pressure.

(Choice B) IPP decreases when the diaphragm contracts. During anesthesia, the diaphragm is paralyzed and cannot decrease the IPP.

(Choice D) During normal inspiration, the alveolar pressure decreases to draw air into the lungs. However, a ventilator forces air into the lungs and increases the alveolar pressure.

Educational objective:
In normal inspiration, the diaphragm contracts to reduce intrapleural pressure, which results in lung expansion. In positive pressure ventilation, an external pump directly increases alveolar pressure by pumping air into the lungs to inflate the lungs.

Time spent:0 sec
Subject:
Physics

QID:400656
System:
Fluids

When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area A and a thickness T is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\text{DA}\ \Delta\ P}{T}$$

Equation 1

where D is the diffusion coefficient specific for the gas and ΔP is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration C of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\text{H}} P_{\text{gas}}$$

Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

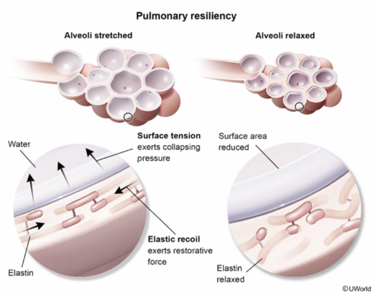
Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

Unassisted expiration is possible under the effects of anesthesia due to:

- A. pressure build-up due to the Venturi effect. [38%]
- B. the elasticity of pulmonary tissue. [44%]
- C. turbulent flow at high positive pressures. [5%]
- D. energy stored in the diaphragm. [10%]

Correct
45% answered correctly

Explanation:



The **resiliency** of the lungs refers to the **lungs' ability to recoil** after being stretched. Pulmonary resiliency has two main components:

- **Elastic recoil** refers to the elasticity of elastin fibers making up alveolar tissues. These fibers are crosslinked and are capable of being stretched like a spring. When stretched, these fibers exert a restorative force on the lungs following inspiration.

- **Surface tension** refers to the tendency of a liquid to reduce exposed surface area due to attractive intermolecular forces between its molecules. Surface tension is significant at alveolar surfaces due to hydrogen bonds between water molecules lining the alveolar sacs. Because surface area increases when the alveoli expand, surface tension exerts a collapsing pressure on the alveoli.

Elastic recoil and surface tension both act to decrease alveolar volume and force air out of the lungs without the use of muscles or ATP energy. Therefore, **passive expiration** is possible even under the paralytic effects of anesthesia.

(Choice A) The Venturi effect refers to the reduction of fluid pressure that occurs when flow velocity increases at constricted sections of a tube. Therefore, the Venturi effect is irrelevant to the pressure in the alveoli and lung exhalation. In fact, constricted airways can cause significant reductions in pressure that cause them to collapse.

(Choice C) Turbulent flow refers to a disorganized type of fluid flow that occurs at high velocities, not high pressures. Turbulent flow increases flow resistance, which would impede exhalation.

(Choice D) Unlike alveolar tissues, the diaphragm is a muscle; it is not made up of elastic fibers. The diaphragm simply relaxes during expiration.

Educational objective:
Pulmonary resiliency (elastic recoil and surface tension effects) refers to the tendency of the lungs to return to their original shape after being stretched during inhalation. Pulmonary resiliency allows for passive expiration and does not require muscles or ATP energy, and can therefore occur under the paralytic effects of anesthesia.

Time spent: 0 sec
Subject:
Physics

QID:400657
System:
Fluids

When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area A and a thickness T is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\text{DA}\ \Delta\ P}{T}$$

Equation 1

where D is the diffusion coefficient specific for the gas and ΔP is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration C of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\text{H}} P_{\text{gas}}$$

Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

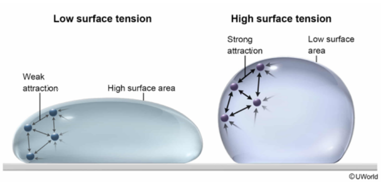
Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

Many inhalational anesthetics are thought to increase surface tension effects in the alveoli. When water has a slightly greater surface tension, it:

- A. forms a convex meniscus in a narrow glass tube. [55%]
- B. evaporates more quickly. [2%]
- C. has weaker van der Waals forces. [2%]
- ✔ D. creates droplets with smaller surface areas. [38%]

Correct
38% answered correctly

Explanation:



The **surface tension** of a liquid refers to the imbalance of intermolecular forces at its surface that makes its surface act as a thin, elastic film. Surface tension arises from **cohesive forces** (between like molecules) in a fluid. Water has relatively high surface tension compared to many liquids due its ability to form hydrogen bonds.

The formation of water droplets is partly due to surface tension effects. Water molecules away from the liquid-air interface experience attractive intermolecular forces from all directions. Conversely, water **molecules at the surface** experience a **net inward force**. This **decreases the surface area** of the water and pulls it into an approximately spherical shape. Because high surface tension indicates high cohesive forces, water with greater surface tension creates droplets with smaller surface areas.

(Choice A) Water in a glass tube forms a concave meniscus, not a convex meniscus. Liquids that form a convex meniscus, such as mercury, have surface tensions about 10 times greater than water.

(Choice B) Evaporation occurs when the kinetic energy of a molecule can overcome intermolecular forces at the surface of the liquid. Fewer evaporative events would occur because more energy would be required to break the greater cohesive forces of high surface tension water.

(Choice C) Van der Waals forces are attractive forces between molecules that do not arise from covalent or ionic bonds. Common examples include dipole-dipole forces and hydrogen bonds. Therefore, water with greater surface tension will have stronger van der

Waals forces.

Educational objective:
The surface tension of a liquid is a result of strong cohesive forces between its molecules. Surface tension creates a tendency to decrease exposed surface area due to a net inward force at the surface molecules.

Time spent: 0 sec
Subject: Physics

QID:400658
System: Fluids

When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area *A* and a thickness *T* is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\text{DA}\,\Delta\,P}{T}$$

Equation 1

where *D* is the diffusion coefficient specific for the gas and ΔP is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration *C* of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\text{H}} P_{\text{gas}}$$

Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

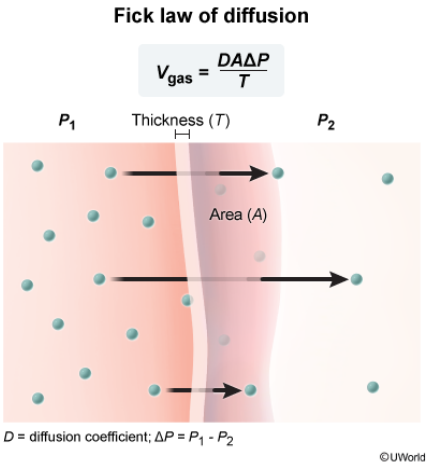
Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

An inhalational anesthetic administered to a patient has a rate of diffusion V_{gas}. If the partial pressure difference is doubled and the membrane thickness is also doubled, what will be the new rate of diffusion?

- A. $V_{gas}/2$ [2%]
- ✔ B. V_{gas} [90%]
- C. $2V_{gas}$ [3%]
- D. $4V_{gas}$ [4%]

Correct
89% answered correctly

Explanation:



The **diffusion of a gas** across a membrane is **driven by partial pressures differences** across the membrane from higher to lower partial pressures. The rate of diffusion of a gas across a membrane with a surface area *A* and a thickness *T* is described by the **Fick law of diffusion** (Equation 1):

$$V_{\text{gas}} = \frac{\text{DA}\,\Delta\,P}{T}$$

where *D* is the diffusion coefficient specific for the gas and Δ*P* is the partial pressure difference across the membrane.

From the above equation, the rate of diffusion is directly proportional to the partial pressure difference Δ*P* and inversely proportional to membrane thickness *T*. If both Δ*P* and *T* are doubled, the **rate of diffusion** will be **unchanged**:

$$\text{new } V_{\text{gas}} = \frac{\text{DA}(2\,\Delta\,P)}{(2T)} = \frac{\text{DA}\,\Delta\,P}{T} = V_{\text{gas}}$$

Educational objective:
The rate of diffusion of a gas across a membrane is described by the Fick law of diffusion: $V_{\text{gas}} = \frac{\text{DA}\,\Delta\,P}{T}$. It is directly proportional to the partial pressure difference and inversely proportional to the thickness of the membrane. Therefore, V_{gas} will remain unchanged if both Δ*P* and *T* are doubled.

Time spent:0 sec
Subject:
Physics

QID:400659
System:
Fluids

When present in the bloodstream above a threshold partial pressure, a general anesthetic blocks the sensation of pain, induces a state of unconsciousness, and causes muscle paralysis. Many anesthetic agents are administered in their gaseous phase and are absorbed into the bloodstream through the lungs. The rate of diffusion V_{gas} of an anesthetic across the alveolar membrane with a surface area A and a thickness T is described by the Fick law of diffusion:

$$V_{\text{gas}} = \frac{\mathrm{DA}\ \Delta\ P}{T}$$

Equation 1

where D is the diffusion coefficient specific for the gas and ΔP is the partial pressure difference across the membrane.

An inhaled anesthetic is described by its *blood-gas partition coefficient*, the ratio of the anesthetic's concentration in the blood to its concentration in the lungs when the partial pressures are equal. The concentration C of an anesthetic gas dissolved in blood is described by the Henry law of solubility:

$$C = k_{\mathrm{H}} P_{\text{gas}}$$

Equation 2

where k_H is the solubility constant specific for the gas and P_{gas} is the partial pressure of the gas.

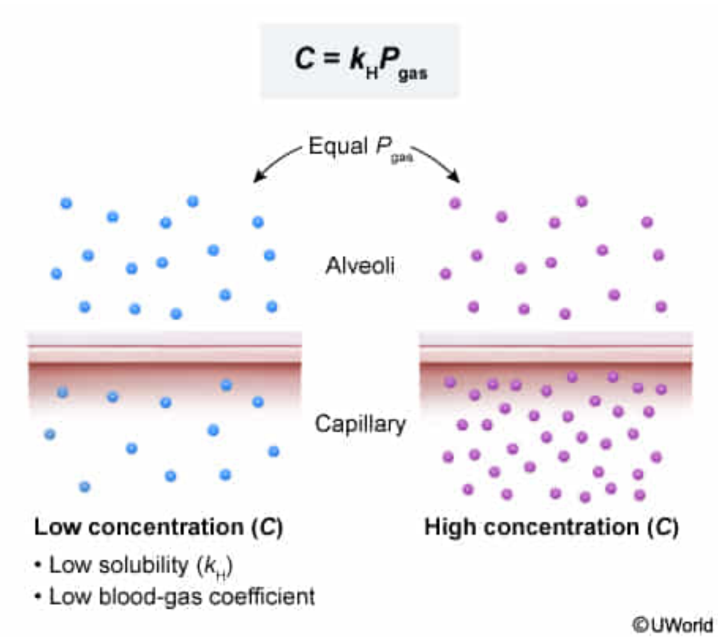
Due to the paralytic effects of general anesthetics, mechanical ventilators are usually necessary to assist a patient's breathing during surgery. Positive-pressure ventilators are noninvasive and use external pumps to induce inspiration by forcing air into the lungs through a ventilation mask. Despite the paralytic effects of general anesthesia, patients are often capable of expiration without the active aid of a mechanical ventilator.

Which of the following characteristics would describe an inhalational anesthetic that is present in a relatively low concentration in the blood when it takes effect?

- ✔ A. Relatively low solubility constant and a low blood-gas coefficient [57%]
- B. Relatively low solubility constant and a high blood-gas coefficient [16%]
- C. Relatively high solubility constant and a low blood-gas coefficient [20%]
- D. Relatively high solubility constant and a high blood-gas coefficient [6%]

Correct
56% answered correctly

Explanation:



According to the **Henry law of solubility** (Equation 2), the concentration C of a gas dissolved in solution is the product of the solubility constant k_H (specific for the gas and solution) and the partial pressure of the gas P_{gas} in its gas phase:

$$C = k_H P_{gas}$$

According to the passage, an anesthetic takes effect when the threshold partial pressure is reached. Rearranging the above equation, the partial pressure of a dissolved gas is inversely proportional to its solubility constant:

$$P_{\text{gas}} = \frac{C}{k_{\text{H}}}$$

In other words, highly soluble anesthetics dissolve easily in the blood, and therefore higher concentrations are required to reach the target threshold partial pressure. In contrast, a **poorly soluble** gas will require a **low concentration** in the blood to take effect.

According to the passage, the **blood-gas coefficient** is the ratio of the anesthetic's concentration in the blood to its concentration in the lungs at equilibrium. If the concentration in the blood is low, the blood-gas coefficient will also be **low**.

(Choices B and D) The blood-gas partition coefficient is directly proportional to the anesthetic's concentration in the blood. Therefore, a poorly soluble gas will have a low concentration in the blood and a low blood-gas partition coefficient.

(Choice C) An inhalational anesthetic with a high solubility constant will have a high concentration in the blood, not a low concentration in the blood.

Educational objective:

The concentration C of a gas dissolved in solution is described by the Henry law of solubility: $C = k_{\text{H}}P_{\text{gas}}$. Therefore, the partial pressure of the dissolved gas is inversely proportional to its solubility constant. In addition, the blood-gas partition coefficient is directly proportional to the concentration in the blood at equilibrium.

Time spent:0 sec
Subject:
Physics

QID:400660
System:
Fluids